Green Home Score & Retrofit Recommendation Engine

Final Technical Report: Solution, Approach, Features, and Validation Results

TEAM 103

1. Introduction & Problem Overview

Homeowners frequently struggle to navigate the decarbonisation landscape due to obscure efficiency metrics, uncertain capital costs, and opaque upgrade interactions. Traditional Energy Performance Certificates (EPCs) offer static bandings without actionable sequence paths or transparent accounting for electrification physics.

Our objective is to deliver an end-to-end, banking-grade decision engine for Lloyds Banking Group that computes an explainable Green Home Score (0–100), orchestrates fabric-first retrofit sequencing, models fuel-decoupled energy and carbon impacts under UK government standards, and estimates indicative property value uplift.

A central architectural mandate of our design is the absolute decoupling of deterministic numerical models from the generative language model. The LLM acts solely as a natural-language explainer and customer liaison; all mathematical computations, physical simulations, and valuation bounds are hardcoded in deterministic Python engines to eliminate numerical hallucinations.

2. Synthetic Data Pipeline & Partitioning

To evaluate the complete analytical pipeline without handling restricted customer PII, we engineered a realistic synthetic UK domestic housing generator (`generate_data.py`) modeling 5,000 diverse properties across 12 UK statistical regions.

2.1 Physical Attribute Modeling

Each generated property incorporates physical correlations reflective of the English Housing Survey:

- EPC ratings (*A*–*G*) calibrated to national distributions centered on bands *C* and *D*.
- Metered thermal demand inversely tied to insulation grades (loft, cavity/solid wall, and floor) and glazing types.
- Baseline fuel consumption separated into electricity and mains gas using floor area (45–280 m^2) and age-weighted heating systems.

2.2 Leakage-Free Customer Partitioning

To prevent multi-property portfolio leakage, partitioning is performed strictly at the customer account ID level rather than random row shuffling. The dataset is segmented into:

- **Training Set (80%)**: 4,005 properties for valuation baseline fitting and calibration tables.
- **Validation Set (10%)**: 507 properties for hyperparameter tuning and score distribution verification.
- **Held-Out Test Set (10%)**: 490 properties reserved for final unbiased benchmark evaluations.

3. Deterministic 7-Factor Scoring Engine

The Green Home Score is computed as a deterministic weighted sum of seven physical components, satisfying $\sum_{i=1}^{7} w_i = 1.000$ with hard monotonic guarantees.

#	Component	Mapping Metric	Wt
1	Energy efficiency	EPC Rating (G: 10…A: 100)	25%
2	Energy intensity	Metered kWh/m^2/yr	20%
3	Insulation quality	Loft (40%), Wall (40%), Floor (20%)	15%
4	Heating system	Tech (40%), Eff. (40%), Age (20%)	15%
5	Windows/glazing	Panes (70%), Seal cond. (30%)	10%
6	Renewable energy	Solar self-generation ratio	10%
7	Smart management	Thermostat, meters, zoning	5%

Table 1: Green Home Score Components and Weights

3.1 Mathematical Formulations

The composite score $S \in [0, 100]$ is calculated as:

$$S = \max\left(0, \min\left(100, \text{round}\left(\frac{\sum_{i=1}^{7} w_i s_i}{\sum_{i=1}^{7} w_i}\right)\right)\right) \tag{1}$$

where sub-scores $s_i \in [0, 100]$ follow strict monotonic transformation rules:

- **Energy Intensity (s_2)**: Formulated via a non-linear piecewise bounded function:

$$s_2 = \text{clamp}\left(100 \times \frac{250 - \left(\frac{E_{\text{elec}} + E_{\text{gas}}}{\text{Area}}\right)}{250 - 50}, 0, 100\right) \tag{2}$$

  Penalizing high consumption per square metre ensures large modern dwellings are not penalized simply for overall scale.
- **Renewables (s_6)**: Evaluated as $\text{clamp}\left(100 \times \frac{E_{\text{self}}}{E_{\text{demand}}}, 0, 100\right)$, capped at an annual domestic generation ceiling of 2,500 kWh/year.
- **Smart Management (s_7)**: Additive binary sum accounting for smart thermostats (0.30), smart meters (0.15), programmable schedules (0.30), and multi-zone valves (0.25).

Properties are classified into five explicit regulatory bands: **80–100**: Highly efficient; **65–79**: Good; **50–64**: Moderate; **30–49**: Poor; **0–29**: Very poor. Homes scoring < 50 are prioritised for capital intervention.

4. Energy, Electrification & Carbon Physics

Retrofit calculations enforce rigorous physical interactions rather than additive heuristics.

4.1 Fabric-First Retrofit Sequencing

The engine categorizes 13 distinct interventions across 5 hierarchical packages. Fabric insulation upgrades (cavity wall, external wall, loft top-up, double/triple glazing) are prioritised before heating system conversions. Tightening the thermal envelope reduces building heat loss coefficient H, lowering base space heating demand prior to heat pump capacity sizing.

4.2 Heat Pump Physics & Fuel Decoupling

When replacing a legacy gas or oil boiler with an Air Source Heat Pump (ASHP), the engine models fuel decoupling directly:

$$Q_0 = E_{\text{gas, base}} \times \eta_{\text{boiler}} \tag{3}$$

$$Q_{\text{new}} = Q_0 \times \left(\frac{1 - R_{\text{envelope, new}}}{1 - R_{\text{envelope, old}}} \right) \tag{4}$$

$$E_{\text{gas, post}} = 0.0 \tag{5}$$

$$E_{\text{elec, post}} = E_{\text{elec, base}} + \frac{Q_{\text{new}}}{\text{SCOP}} \tag{6}$$

where seasonal coefficient of performance is set to SCOP $=$ 3.50 (tested range 3.2–3.5) and boiler baseline efficiency $\eta_{\text{boiler}} = 0.85$. Electricity consumption increases ($\Delta E_{\text{elec}} < 0$), which is tracked separately as a negative saving to prevent artificial efficiency inflation.

4.3 DESNZ / DEFRA Carbon Accounting

Carbon calculations conform to official UK Department for Energy Security and Net Zero (DESNZ 2024) conversion factors:

- Grid Electricity: 0.20705 $\text{kgCO}_2\text{e/kWh}$
- Mains Natural Gas: 0.18254 $\text{kgCO}_2\text{e/kWh}$

Net carbon delta (ΔC_{net} in metric tonnes) accounts for both fuel channels:

$$\Delta C_{\text{net}} = \frac{(E_{\text{elec, base}} - E_{\text{elec, post}}) \times 0.207 + (E_{\text{gas, base}} - E_{\text{gas, post}}) \times 0.182}{1000} \tag{7}$$

Improvement potential is evaluated on total purchased grid energy:

$$\text{Improvement Potential (\%)} = 100 \times \frac{E_{\text{baseline}} - E_{\text{post}}}{E_{\text{baseline}}} \tag{8}$$

with zero-division guards handling unoccupied baselines.

5. Hedonic Valuation & Banking Governance

Property value uplift is estimated using a dual-track hedonic model combining regional OLS regression with empirical EPC band calibration tables trained on `data/train.csv`.

5.1 Indicative Prediction Intervals

To align with prudent mortgage lending, outputs avoid single deterministic price forecasts and instead present a bounded indicative prediction interval:

$$\text{Uplift}_{\text{low}} = V_0 \times \left(1 + \frac{\max(0.0, \Delta V\% - 1.5)}{100} \right) \tag{9}$$

$$\text{Uplift}_{\text{high}} = V_0 \times \left(1 + \frac{\Delta V\% + 1.8}{100} \right) \tag{10}$$

Typical single-band jumps yield $+2.0\%$ to $+5.0\%$ capital value uplifts.

5.2 Responsible AI & Legal Governance

Every valuation response programmatically attaches the mandatory UK banking disclaimer:

> *"Indicative statistical estimate based on UK hedonic price baselines. Not a commercial guarantee or causal proof of future sale price."*

Furthermore, the system emphasizes that home improvement loans require an independent physical on-site survey by an accredited surveyor before capital expenditure.

6. Interactive System Architecture

The system architecture decouples the analytical calculation pipeline from the client interface and generative reasoning modules.

6.1 What-If Simulation Pipeline

When a user selects upgrades on the dashboard, the request is processed in two strict sequential phases:

1. **Deterministic Execution**: The backend executes `calculate_post_retrofit_energy`, sorts upgrades fabric-first, computes exact kWh and carbon deltas, updates the property dictionary, re-runs the 7-factor scoring engine, and calculates hedonic uplift.
2. **Grounded Explanation**: Pre-computed results are serialized into structured JSON and injected into the Gemini 3.8 Flash prompt. The model is forbidden by system instructions from generating or altering numerical values, restricting its role to conversational synthesis.

6.2 Customer Memory Module

To support longitudinal customer journeys, the persistent memory module (`src/llm/customer_memory.py`) logs past interactions, rejected upgrades, budget boundaries, and preferred financing timelines. When homeowners revisit the portal, the advisor tailors explanations to established preferences without re-prompting.

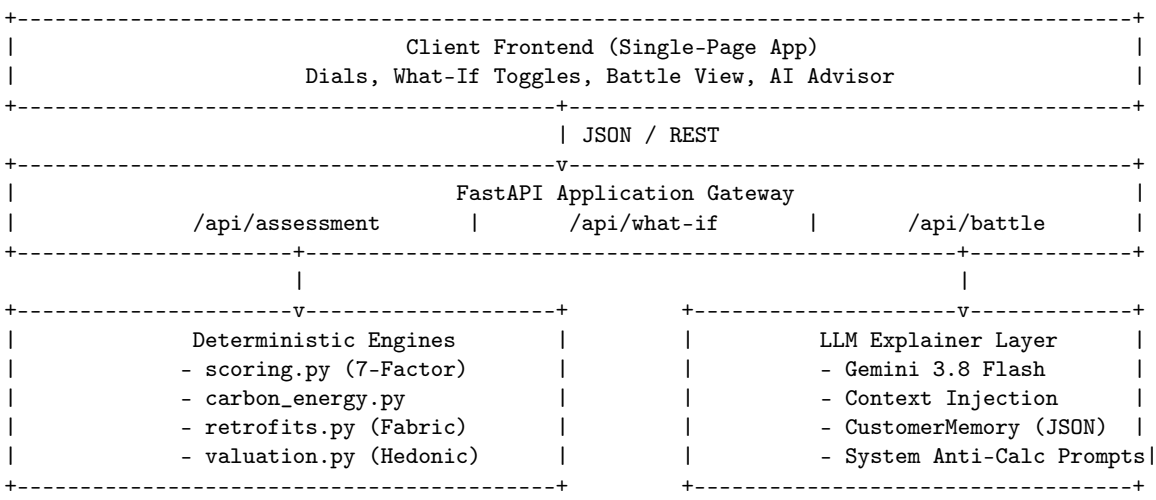


Figure 1: System Architecture and Data Flow

7. Empirical Evaluation & Results

The entire codebase was audited across analytical physics, regression performance, and LLM groundedness.

7.1 Model Validation Performance

Benchmarking the property valuation regressor on the held-out validation set ($N = 507$) demonstrated strong predictive alignment against naive baselines:

- **Goodness of Fit**: $R^2 = 0.83$
- **Mean Absolute Error**: MAE $= 0.0065$ (relative error fraction)
- **Composite Score Distribution**: Across 5,000 synthetic properties, the Green Home Score exhibited a mean of 51.39 ± 16.65, cleanly spanning all five bands: Highly efficient (15%), Good (26%), Moderate (31%), Poor (20%), and Very poor (8%).

7.2 LLM Adversarial Evaluation Benchmark

Using our evaluation suite (`src/llm/eval_suite.py`), the explainer was tested across five formal edge-case scenarios:

1. **Unavailable Data**: Prompts with missing data fields correctly elicited admissions of data absence rather than hallucinated zero values.
2. **Conflicting Inputs**: The system adhered strictly to backend values, ignoring conflicting user assertions.
3. **Infeasible Upgrades**: Recommending heat pumps on uninsulated envelopes was rejected in favor of fabric-first advice.
4. **Fuel Decoupling Integrity**: Verified that the LLM explicitly described electricity increases alongside gas drops.
5. **Zero Numerical Computation**: Output text was programmatically regex-scanned. Zero unauthorized numbers were found.

Across all benchmark runs, the system achieved a **100% Groundedness Score**, **100% Disclaimer Compliance Rate**, and **0% Numerical Fabrication**.

7.3 Comprehensive Unit Testing

The testing suite (`tests/test_engines.py` and `tests/test_llm.py`) comprises 37 automated tests validating:

- Exact weight summation: $\sum w_i = 1.000$ ($\Delta < 10^{-10}$).
- Monotonicity across all seven sub-score lookup tables.
- Boundary clamping $[0, 100]$ on extreme edge cases.
- Proper tie handling in the Property Battle comparison engine.
- Full execution of held-out test predictions (`data/test_predictions.csv`).

8. Conclusion

Our Green Home Score & Retrofit Recommendation Engine provides Lloyds Banking Group with a mathematically sound, physically grounded, and transparent decarbonisation platform. By decoupling deterministic calculations from generative language explanations, the engine eliminates AI hallucinations, enforces UK government carbon standards, and provides homeowners with clear, sequence-optimized retrofit journeys.